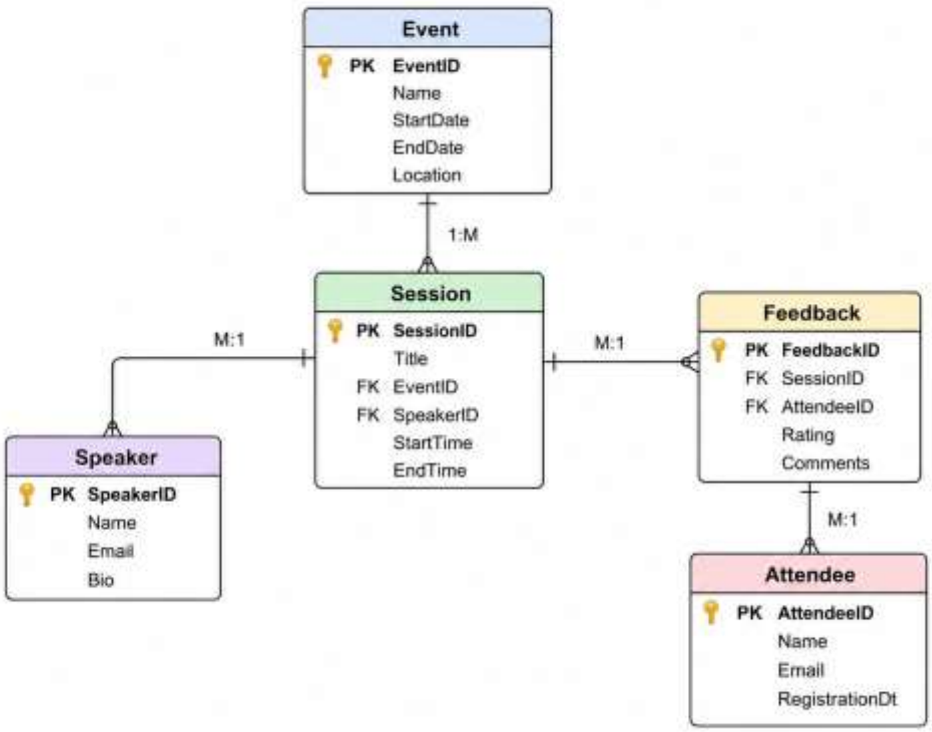
EVENT MANAGEMENT SYSTEM

PART 1

1. **Draw an ER diagram showing objects, primary keys, foreign keys, and relationships (1:M, M:M).**



2. **Identify which relationships should be Lookup and which should be Master-Detail, justifying your choices.**

Event → Session: Master-Detail Relationship
Reason:
A Session cannot exist without an Event. If the parent Event is deleted, related Sessions should also be deleted.

Speaker → Session: Lookup Relationship
Reason:
A Speaker can exist independently even if no Session is assigned.

Session → Feedback: Lookup Relationship
Reason:
Feedback records are associated with Sessions but can be managed independently.

Attendee → Feedback: Lookup Relationship
Reason:
Attendees can exist even if they have not submitted any Feedback.

3. **Salesforce Schema Builder**

Schema Builder is a graphical tool used to visualize and manage the Salesforce data model.

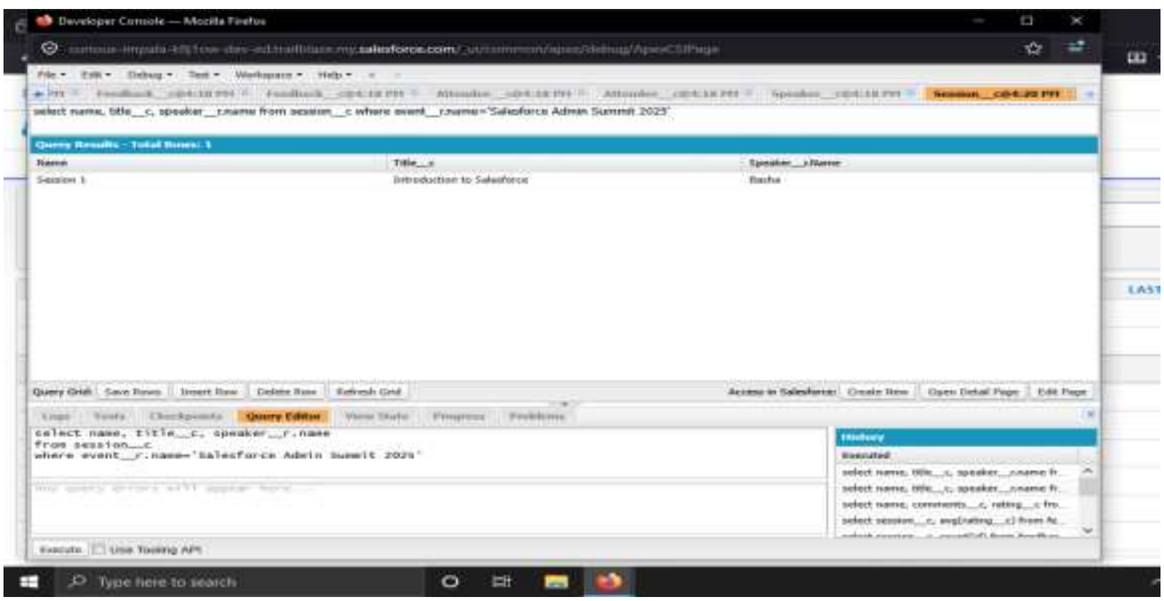
Uses:
- Create custom objects
- Create custom fields
- Create relationships
- Visualize object connections
- Modify existing schema
- Understand parent-child relationship

PART 2: SOQL QUERY TASKS

1. **List all sessions for the event "Salesforce Admin Summit 2025" with their speakers**

SELECT Name, Title__c, Speaker__r.Name
FROM Session__c
WHERE Event__r.Name = 'Salesforce Admin Summit 2025'

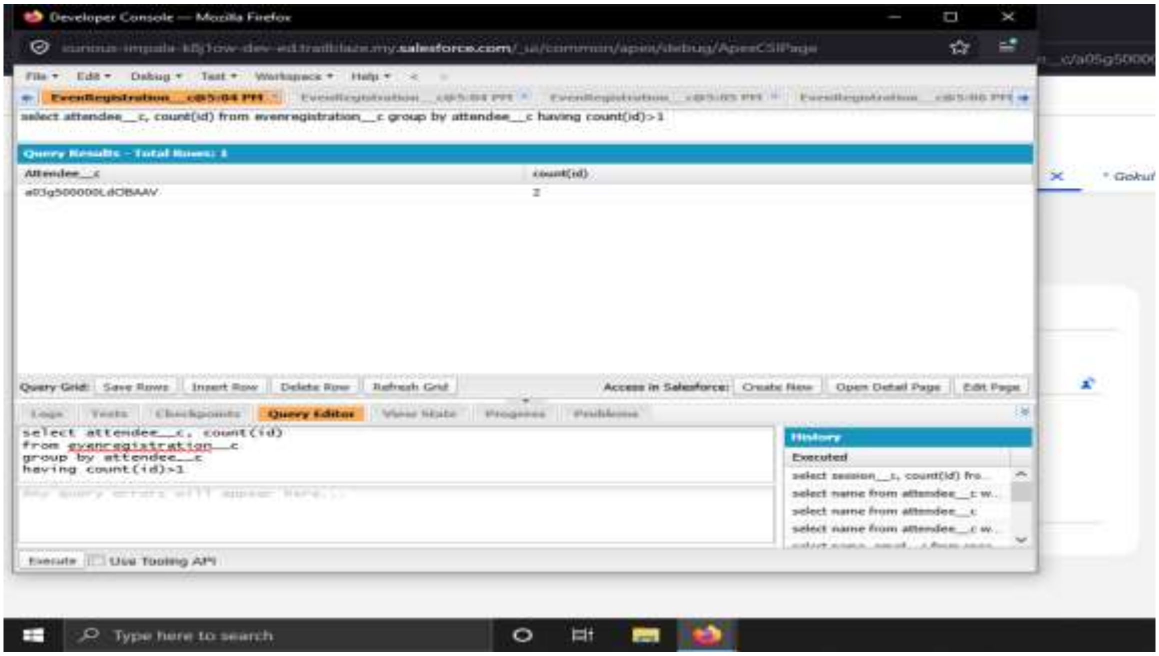
Output:



2. **Find all attendees who have registered for more than one event**

SELECT Attendee__c, COUNT(Id)
FROM EventRegistration__c
GROUP BY Attendee__c
HAVING COUNT(Id) > 1

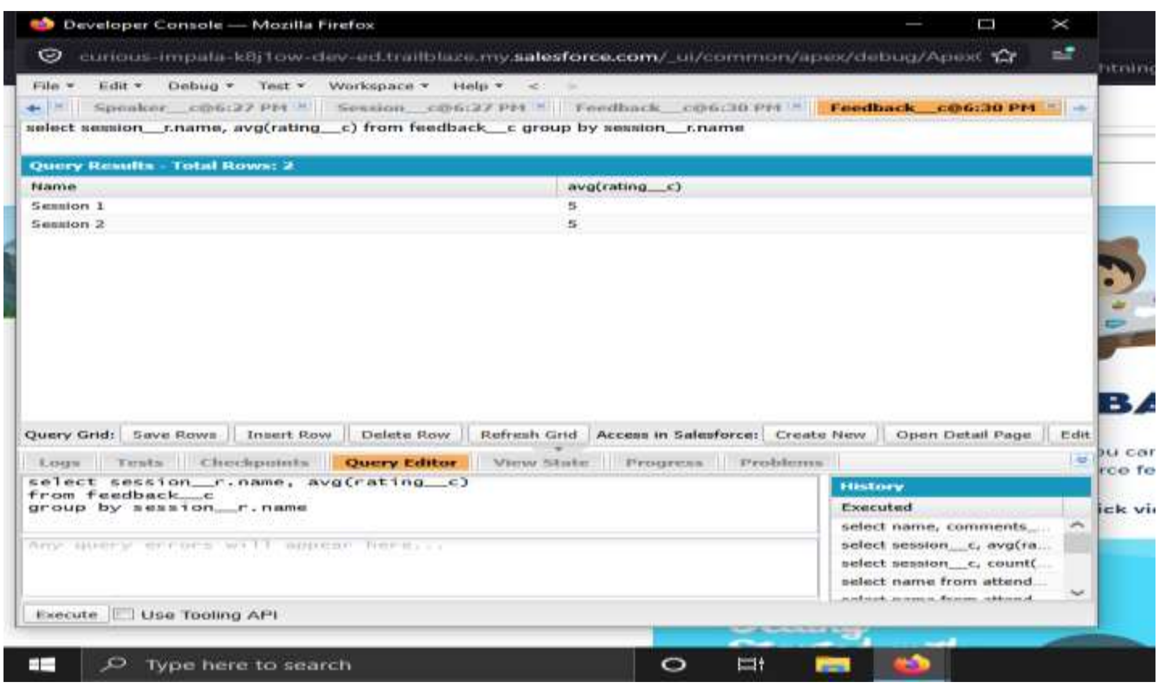
Output:



3. For each session, show the average feedback rating

SELECT Session__r.name, AVG(Rating__c)
FROM Feedback__c
GROUP BY Session__r.name

Output:

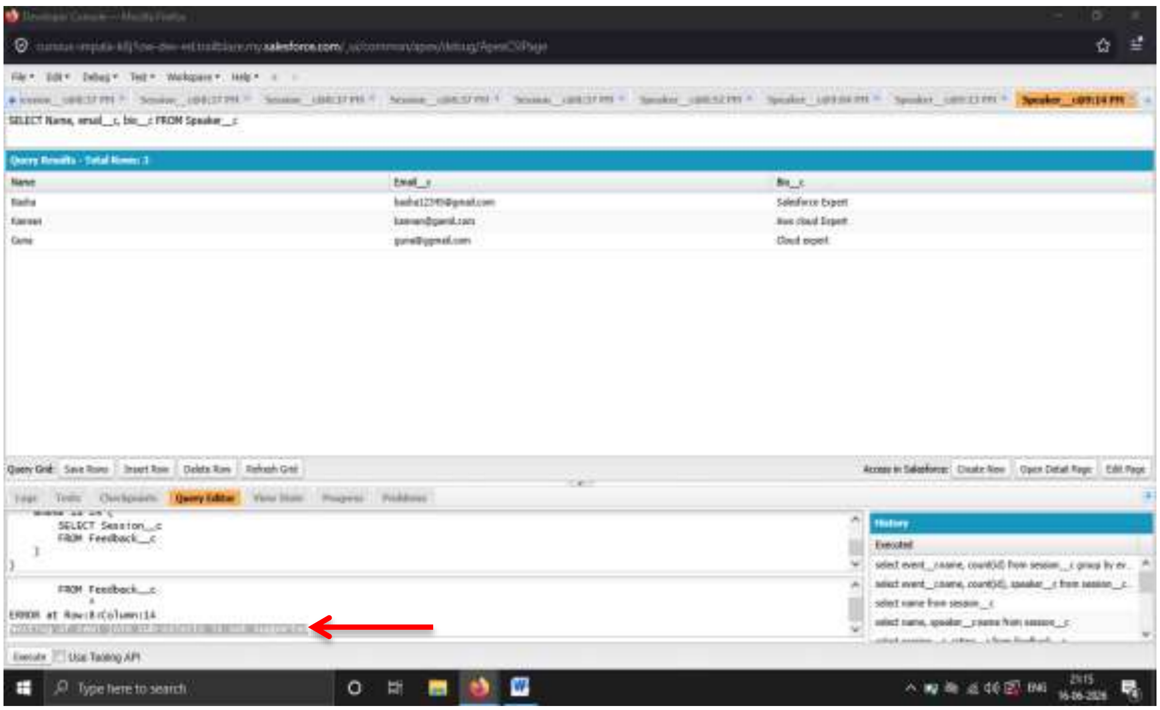


4. List all speakers who have not received any feedback for their sessions

```
SELECT Id, Name
 FROM Speaker__c
 WHERE Id NOT IN (
 SELECT Speaker__c
 FROM Session__c
 WHERE Id IN (
 SELECT Session__c
 FROM Feedback__c
 )
 )
```

Output:

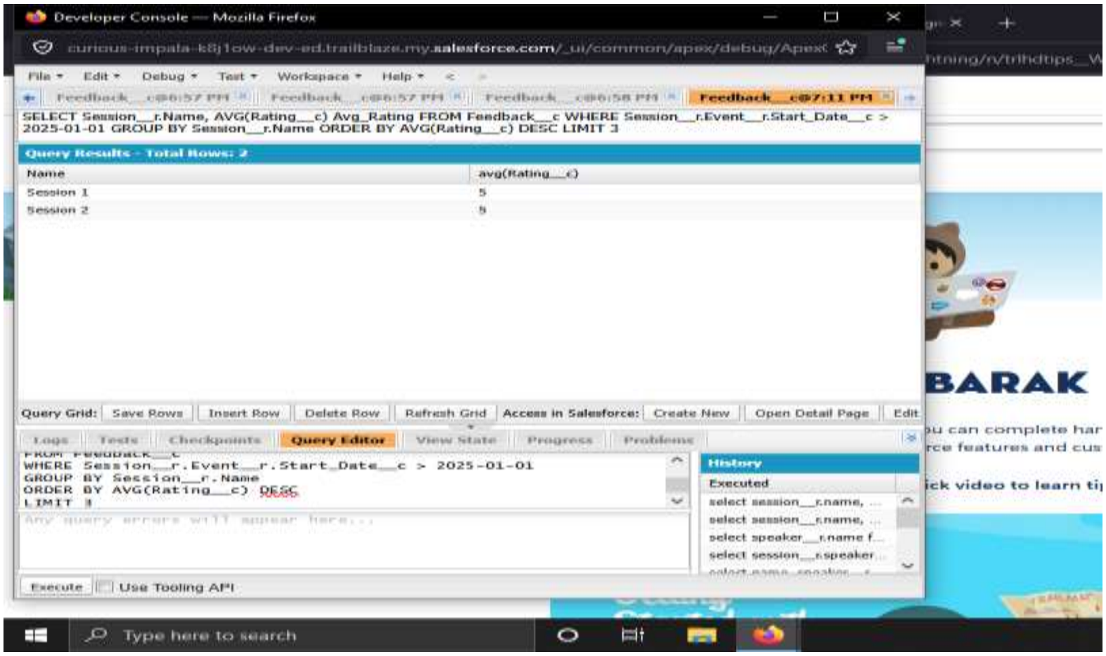
Nesting of semi join sub-selects is not supported



5. Show the top 3 sessions (by average rating) for events held after January 1, 2025

```
SELECT Session__r.Name,
 AVG(Rating__c) Avg_Rating
 FROM Feedback__c
 WHERE Session__r.Event__r.Start_Date__c > 2025-01-01
 GROUP BY Session__r.Name
 ORDER BY AVG(Rating__c) DESC
 LIMIT 3
```

Output:



PART 3: SALESFORCE CONCEPTUAL QUESTIONS

1. Difference Between Lookup and Master-Detail Relationships

Lookup Relationship:
- Loose relationship between objects.
- Child records can exist independently.
- Parent and child have separate ownership.

Example:
Speaker -> Session

Master-Detail Relationship:
- Strong parent-child relationship.
- Child records depend on the parent.
- Child inherits ownership and security settings from the parent.

Example:
Event -> Session

2. Governor Limits

Governor Limits are restrictions enforced by Salesforce to prevent excessive resource consumption in a multi-tenant environment.

Importance:
- Ensures fair usage
- Prevents system overload
- Maintains platform performance

3. Data Quality and Integrity

I would enforce data quality and integrity by making important fields required so users cannot leave them empty. I would use validation rules to prevent incorrect data entry, such as checking valid dates, ratings, and formats. I would apply unique constraints for fields like email IDs to avoid duplicate records.

Examples:
- Event Name must be mandatory.
- End Date must be greater than Start Date.
- Email fields should be unique.
- Rating should be between 1 and 5.

4. Purpose of Schema Builder

Schema Builder helps administrators:
- Visualize data models
- Create objects and fields
- Manage relationships
- Understand dependencies
- Modify schema using a drag-and-drop interface

PART 4: BONUS (CONCEPTUAL)

A conceptual Apex Trigger can automatically mark a Session as "Full" when the number of related records reaches a predefined capacity.

Logic:

1. Count Records associated with a Session.
2. Compare the count with the Session Capacity field.
3. If count equals or exceeds Capacity:
   - Update Session Status to "Full".
4. Otherwise:
   - Keep Status as "Available".